

1. Data Wrangling Viva Questions

Q1. What is Data Wrangling?

Data wrangling is the process of cleaning, transforming, and organizing raw data into useful format for analysis.

Q2. Why is data preprocessing important?

- Improves data quality
- Removes missing values and errors
- Makes data suitable for machine learning
- Increases accuracy of analysis

Q3. What is Pandas?

Pandas is a Python library used for data manipulation and analysis using DataFrames.

Q4. What is a DataFrame?

A DataFrame is a two-dimensional table-like structure in Pandas having rows and columns.

Q5. What is the use of isnull()?

isnull() is used to check missing values in the dataset.

Q6. What is normalization?

Normalization scales numeric values into a fixed range to improve model performance.

Q7. Difference between structured and unstructured data?

- Structured data → Organized in rows and columns
- Unstructured data → Text, images, videos without fixed format

Q8. What are categorical variables?

Variables containing categories or labels like gender, city, department etc.

Q9. How do you convert categorical data into numeric data?

Using encoding techniques like Label Encoding or One-Hot Encoding.

Q10. What is missing data?

Values that are not available in the dataset are called missing data.

2. Outliers and Transformations Viva

Q11. What is an outlier?

An outlier is a data value that is very different from other values.

Q12. How do you detect outliers?

Using:

- Box plot
- IQR method
- Z-score method

Q13. What is data transformation?

Changing the format or scale of data for better analysis is called data transformation.

Q14. Why is skewness reduced?

To make data distribution more normal and improve model performance.

Q15. What is standardization?

Standardization converts data into a form with mean 0 and standard deviation 1.

3. Descriptive Statistics Viva

Q16. What is mean?

Mean is the average of all values.

Q17. What is median?

Median is the middle value after arranging data in order.

Q18. What is mode?

Mode is the most frequently occurring value.

Q19. What is standard deviation?

Standard deviation measures how much data values vary from the mean.

Q20. Difference between variance and standard deviation?

- Variance = square of deviation
- Standard deviation = square root of variance

Q21. What is percentile?

Percentile indicates percentage of values below a particular value.

4. Linear Regression Viva**Q22. What is Linear Regression?**

Linear Regression is a supervised learning algorithm used for prediction of continuous values.

Q23. What is dependent variable?

The output variable to be predicted is called dependent variable.

Q24. What is independent variable?

Input features used for prediction are called independent variables.

Q25. What is the equation of linear regression?

$$y = mx + c$$

Where:

- y = predicted value
- m = slope
- c = intercept

Q26. What is overfitting?

When model performs well on training data but poorly on test data.

Q27. What is train-test split?

Dividing dataset into training and testing sets for model evaluation.

5. Logistic Regression Viva**Q28. What is Logistic Regression?**

Logistic Regression is a classification algorithm used for binary outcomes like Yes/No.

Q29. Difference between Linear and Logistic Regression?

- Linear Regression → Predicts continuous values
- Logistic Regression → Predicts categories/classes

Q30. What is sigmoid function?

Sigmoid function converts values into probability between 0 and 1.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Q31. What is confusion matrix

A table used to evaluate classification model performance.

Q32. What is accuracy?

Accuracy is percentage of correctly predicted values.

Q33. What is precision?

Precision measures correctness of positive predictions.

Q34. What is recall?

Recall measures how many actual positives are correctly predicted.

6. Naïve Bayes Viva

Q35. What is Naïve Bayes algorithm?

Naïve Bayes is a probabilistic classification algorithm based on Bayes theorem.

Q36. Why is it called “Naïve”?

Because it assumes all features are independent.

Q37. What is Bayes theorem?

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B | A)$$

$$P(B | \neg A)$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \approx 0.68, P(B) \approx 0.25$$

$P(B)=0.25$
 $P(B|A)P(A)=0.17$
 $P(A|B)\sim 0.68$
Posterior = useful evidence / total evidence

Q38. Advantages of Naïve Bayes?

- Fast
- Simple
- Works well on text data

7. Text Analytics Viva

Q39. What is NLP?

Natural Language Processing is a field that helps computers understand human language.

Q40. What is tokenization?

Breaking text into smaller parts called tokens.

Q41. What are stop words?

Common words like “is”, “the”, “and” removed during preprocessing.

Q42. What is stemming

Reducing words to root form.

Example: playing → play

Q43. What is lemmatization?

Converting words into meaningful base words.

Q44. What is TF-IDF?

TF-IDF measures importance of words in a document.

8. Data Visualization Viva**Q45. Why is data visualization important?**

It helps understand patterns, trends, and relationships easily.

Q46. What is a histogram?

Histogram shows frequency distribution of continuous data.

Q47. What is box plot?

Box plot shows distribution and outliers in data.

Q48. What is Seaborn?

Seaborn is a Python visualization library built on Matplotlib.

Q49. Difference between bar chart and histogram?

- Bar chart → categorical data
- Histogram → continuous data

9. Apache Spark Viva

Q50. What is Apache Spark?

Apache Spark is a fast big data processing framework.

Q51. What is Scala?

Scala is a programming language used with Apache Spark.

Q52. Why Spark is faster than Hadoop?

Because Spark processes data in memory.

10. Recommendation System Viva**Q53. What is recommendation system?**

A system that suggests products or movies to users based on preferences.

Q54. What is collaborative filtering?

Recommendation based on behavior of similar users.

Q55. What is content-based filtering?

Recommendation based on item features.

Common General Viva Questions**Q56. Why did you use Python?**

Because Python is simple, powerful, and has many libraries for data science.

Q57. What libraries did you use?

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Q58. What is machine learning?

Machine learning enables computers to learn from data and make predictions.

Q59. Difference between supervised and unsupervised learning?

- Supervised → data with labels
- Unsupervised → data without labels

Q60. What is overfitting and underfitting?

- Overfitting → model learns too much from training data
- Underfitting → model fails to learn properly